Laboratory Tasks

Task 1: Sum Of Natural Numbers

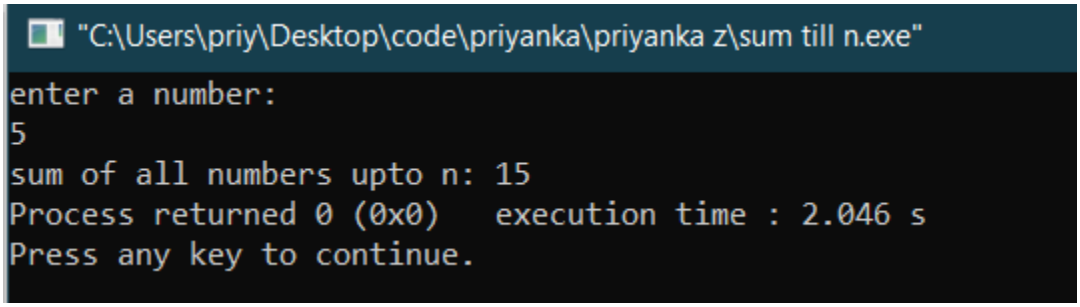
```c
#include <stdio.h>

int main () {
    int n;
    printf("enter a number: \n");
    scanf("%d", &n);
    int s = 0;

    for (int i = 0; i <= n; i++) {
        s = s + i;
    }

    printf("sum of all numbers upto n: %d", s);
    return 0;


}
```



Task 2: Prime Number Check

```c
#include <stdio.h>

int main() {
    int n;

    printf("enter a number: ");
    scanf("%d", &n);

    if (n < 2) {
        printf("%d is not a prime number\n", n);
        return 0;
    }

    for (int i = 2; i <= n / 2; i++) {
        if (n % i == 0) {
            printf("%d is not a prime number\n", n);
            return 0;
        }
    }

    printf("%d is a prime number\n", n);

    return 0;
}
```

```
enter a number: 7
7 is a prime number

Process returned 0 (0x0)   execution time : 2.228 s
Press any key to continue.
```

Task 3: Fibonacci Series with Analysis

```c
#include <stdio.h>

int main() {
    int n;
    printf("enter the number of terms: \n");
    scanf("%d", &n);

    int a = 0;
    int b = 1;
    int k;

    int i = 0;
    while (i < n) {
        printf("%d ", a);
        k = a + b;
        a = b;
        b = k;
        i++;
    }

    return 0;
}
```

```
enter the number of terms:
10
0 1 1 2 3 5 8 13 21 34
Process returned 0 (0x0)   execution time : 1.385 s
Press any key to continue.
```

Additional Practice Tasks

Task 1: Reverse A Number

```c
#include <stdio.h>

int main() {
int n;

printf("enter the number you want to reverse: ");
scanf("%d", &n);

int d = 0; int a = 0;

while (n != 0) {
d = n % 10;
a = a * 10 + d;
n = n / 10;
}

printf("the reversed number is : %d", a);
return 0;


}

```

```
enter the number you want to reverse: 2383
the reversed number is : 3832
Process returned 0 (0x0)   execution time : 1.586 s
Press any key to continue.
```

Task 2: Factorial Of A Number

```c
#include <stdio.h>

int main () {
int n;
printf("enter a number: \n");
scanf("%d", &n);

int a = 1;

for (int i = 1; i <= n; i++){;
a = a * i;
}

printf("the factorial of the number is %d", a);
return 0;
}
```

```
enter a number:
7
the factorial of the number is 5040
Process returned 0 (0x0)   execution time : 1.731 s
Press any key to continue.
```

Task 3: Print Multiplication Table

```c
#include <stdio.h>

int main () {

int n;
printf("enter a number: \n");
scanf("%d", &n);

for (int i = 1; i <= 10; i++) {
printf("%d ", n * i);
}

return 0;
}

```

```
enter a number:
24
24 48 72 96 120 144 168 192 216 240
Process returned 0 (0x0)   execution time : 0.791 s
Press any key to continue.
```

Task 4: Count Number Of Digits

```c
#include <stdio.h>

int main () {
int n;
printf("enter a number: ");
scanf("%d", &n);

int d = 0; int a = 0;

while (n != 0) {
d = n % 10;
a++;
n = n / 10;
}

printf("the number of digits is: %d", a);
return 0;
}
```

```
enter a number: 26738
the number of digits is: 5
Process returned 0 (0x0)   execution time : 1.493 s
Press any key to continue.
```

Task 5: Right Angled Triangle

```c
#include <stdio.h>

int main () {

int n;
printf ("enter number of rows: ");
scanf("%d", &n);

for (int i = 1; i <= n; i++) {
for (int j = 1; j <= i; j++) {
printf("* ");
}
printf("\n");
}
return 0;
}
```

```
enter number of rows: 5
*
* *
* * *
* * * *
* * * * *

Process returned 0 (0x0)   execution time : 0.761 s
Press any key to continue.
```

Ex. 3

Q.1)
For: Initialisation, condition and updates are in the loop header.
While: Condition is in the header, the initialisation and update are separately written.

Q.2) In a do-while loop, the body executes before the condition is checked, so it runs at least once.

Q.3) An infinite loop never ends.
eg) while (1) {
    printf("hello");
}

Q.4)
Break: exits the loop completely.
Continue: skips the current iteration and moves to the next.

Q.5)
```
int i=1;
while (i<=n) {
    sum = sum + i;
    i++;
}
```

Q.6)
Nesting: Used when one loop must repeat inside another.
eg) Bubble sort.

Q.7) The loop may become an infinite loop because the condition never changes.

Q.8) 1 2 4 5

Q.9) 1 2 3

Q.10) 10